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Data Structures and Algorithms

DSA101 | Suggested sequence · Semester 2 | Student-authored study guide

How to use these notes

Complexity, arrays, linked structures, stacks, queues, hash maps, trees, graphs, and sorting. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Complexity and linear structures

Describe growth as input size grows. O(1) is constant, O(log n) logarithmic, O(n) linear, and O(n log n) common for efficient comparison sorting. Arrays give indexed access; linked lists trade that for pointer-based traversal.

Worked example

```
One pass to find a maximum is O(n) time and O(1) extra space.
```

Check your understanding

- What does asymptotic notation ignore?
- Why is array indexing O(1)?

MODULE 02 · Stacks, queues, and hashing

A stack is last-in/first-out; a queue is first-in/first-out. A hash map supports expected constant-time lookup when keys distribute well, but collisions must be handled and worst-case behavior can be slower.

Worked example

```
Use a stack to validate brackets: push opening symbols; each closing symbol must match the top.
```

Check your understanding

- Which structure models a printer queue?
- What is a hash collision?

MODULE 03 · Trees, graphs, and sorting

Trees model hierarchy; binary search trees depend on an ordering invariant. Graphs model relationships; BFS explores by distance layers and DFS explores paths. Compare algorithms by constraints, correctness, and memory.

Worked example

```
BFS with a queue finds shortest paths in an unweighted graph.
```

Check your understanding

- Why does BFS find unweighted shortest paths?

• What precondition allows binary search?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.